Name : Sanket Gorade | HBM25062

1: Find Two Missing Frequencies using Mean (26) and Total Frequency

Class interval	Frequency (f)	midpoint (m)	f*m
0 - 10	10	5	50
10 - 20	x	15	15x
20 - 30	25	25	625
30 - 40	y	35	35y
40 - 50	10	45	450
Total	80	—	—

$\Sigma f = 10 + x + 25 + y + 10 = 80$

$\therefore x + y = 80 - 45 = 35$

$\therefore x + y = 35$ —— ①

Now,

$$\bar{x} = \frac{\Sigma f * m}{\Sigma f} = \frac{15x + 35y + 1125}{80}$$

$$\therefore 26 = \frac{15x + 35y + 1125}{80}$$

$\therefore 15x + 35y = 955$ —— ②

Solving eqn ① & ②

we get,

$$\boxed{\begin{array}{l} x = 13.5 \\ y = 21.5 \end{array}}$$

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2: Find Two Missing Frequencies using Median and Total Frequency

Class Interval	Frequency (f)	Cumulative freq. (CF)
0 - 20	5	5
20 - 40	x	$5+x$
40 - 60	30	$5+x+30 = 35+x$
60 - 80	y	$35+x+y$
80 - 100	10	$35+x+y+10 = 45+x+y$
Total	70	70

← median class

The Median of the distribution is 46.

Also calculate Q3, P77

→ Here,

$70 = 45 + x + y$

$\therefore x + y = 25$ — (1)

Now,

$\text{Median} = L + \left(\frac{N/2 - CF}{f}\right) * h$

where, L = lower boundary of class $= 40$

→ $N/2$ = Half of total frequency $= \frac{70}{2} = 35$

→ CF = Cumulative frequency of class preceding median class $= 5+x$

→ f = freq. of median class $= 30$

& → h = class size $= 60-40 = 20$.

$\therefore \text{Median} = 40 + \left(\frac{35 - (5+x)}{30}\right) * 20$

$\therefore 46 = 40 + \left(\frac{30-x}{3\cancel{0}}\right) * 2\cancel{0}$

$\therefore \frac{\cancel{46} \times 30}{\cancel{20}}$ $\therefore 6 \times \frac{30}{20} = 30 - x$

$\therefore 9 = 30 - x$

$\boxed{\therefore x = 21}$

Now, from eq^n (1)

$y = 25 - x = 25 - 21 = 4$

$\boxed{\therefore y = 4}$

Q.2

→ To find Q_3 & P_{77}

Class	f	CF
0-20	5	5
20-40	$\boxed{21}$	26
40-60	30	56
60-80	$\boxed{4}$	60
80-100	10	70
	N=70	

→ formula for quartile or percentile

$$\text{Value} = L + \left(\frac{\text{Position} - CF}{f}\right) * h$$

① To calculate quartile (Q_3)

→ $Q_3 = \frac{3N}{4} = \frac{3 \times 70}{4} = 52.5$

→ 52.5 falls in 40-60 class

⇒ since, $CF = 26 < 52.5 < 56$

∴ $L = 40$, $cf = 26$ (cf of preceding class),

∴ $f = 30$ & $h = 20$.

$\therefore Q_3 = 40 + \left(\frac{52.5 - 26}{30}\right) * 20$

$\boxed{\therefore Q_3 \approx 57.67}$

② To calculate 77^{th} percentile (P_{77})

→ $P_{77} = \frac{77 * N}{100} = \frac{77 \times 70}{100} = 53.9$

→ 53.9 falls in 40-60 class,

∴ $L = 40$, $CF = 26$, $f = 30$, $h = 20$

$\therefore P_{77} = 40 + \left(\frac{53.9 - 26}{30}\right) * 20$

$\boxed{\therefore P_{77} \approx 58.6}$

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3. The table below shows the distribution of monthly wages of 50 workers. Calculate the Standard Deviation .

Wages (₹)	No. of Workers (f)	midpoint (m)	$d = \frac{m-A}{h}$	fd	fd^2
200 - 300	5	250	-3	-15	45
300 - 400	8	350	-2	-16	32
400 - 500	12	450	-1	-12	12
500 - 600	15	550	0	0	0
600 - 700	10	650	1	10	10
	$\Sigma f = 50$			$\Sigma fd = -33$	$\Sigma fd^2 = 99$

→ here, let's select midpoint of 500-600 class as the assumed mean (A) = 550

→ class size = h = 600 - 500 = 100

→ Now, put value of A & h in above table.

→ standard deviation $= \sigma = h * \sqrt{\frac{\Sigma fd^2}{\Sigma f} - \left(\frac{\Sigma fd}{\Sigma f}\right)^2}$

$$= 100 * \sqrt{\frac{99}{50} - \left(\frac{-33}{50}\right)^2}$$

$$\sigma = 100 * \sqrt{1.98 - 0.4356}$$

$$\therefore \sigma = 100 * 1.2427$$

$$\therefore \sigma \approx 124.27$$

∴ The standard deviation of monthly wages is approx. Rs. 124.27.

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4. A manufacturing company wants to compare the consistency of daily output between two production plants, Plant A and Plant B. The data for the daily output (in units) over the last month are summarized below:

Statistic	Plant A (Group 1)	Plant B (Group 2)
Number of Days (N)	30	30
Average Daily Output ($\bar{X}$)	250 units	280 units
Standard Deviation (σ)	25 units	30 units

Questions:

1. Calculate the Coefficient of Variation (CV) for both Plant A and Plant B.
2. Determine which plant shows greater consistency (less variability) in its daily output.

→ CV measures the relative variability of data, expressed as a percentage.

$$\therefore CV = \frac{\text{std. deviation } (\sigma)}{\text{mean } (\bar{x})} * 100$$

(Q.1) →

(1) To find $(CV)_A$:

Avg. daily output $(\bar{X}_A)$ = 250 units

Std. deviation (σ_A) = 25 units

$$\therefore CV_A = \left(\frac{\sigma}{\bar{X}}\right)_A * 100$$

$$\therefore CV_A = \frac{25}{250} * 100 = 10\% \Rightarrow \boxed{\therefore CV_A = 10\%}$$

(2) To find CV_B:

Avg. daily o/p $(\bar{X}_B)$ = 280 units

Std. deviation (σ_B) = 30 units

$$\therefore CV_B = \left(\frac{\sigma}{\bar{X}}\right)_B * 100$$

$$= \frac{30}{280} * 100 = 10.71\% \Rightarrow \boxed{CV_B = 10.71\%}$$

(Q.2) →
Since CV_A (10.00%) is less than CV_B (10.71%), plant A shows greater consistency (less variability) in its daily output compared to plant B.

HBM 25062

5. A recruitment agency wants to assess the relationship between the score a candidate receives on a Pre-Interview Aptitude Test (X) and their eventual First-Year Job Performance Rating (Y) out of 100. They collected data for five candidates.

Candidate	Aptitude Test Score (X)	Job Performance Rating (Y)	X^2	Y^2	XY
1	65	70	4225	4900	4550
2	75	78	5625	6084	5850
3	80	85	6400	7225	6800
4	55	60	3025	3600	3300
5	90	92	8100	8464	8280

here,
$\Sigma X = 365$
$\Sigma Y = 385$
$\Sigma X^2 = 27375$
$\Sigma Y^2 = 30273$
$\Sigma XY = 28780$

What does this result suggest about the usefulness of the aptitude test for predicting job performance?

→ Here, To access the relationship between the score & rating, it can be done by calculating correlation coefficient (r), which quantifies the strength & direction of linear relationship between two variables.

Here, The pearson correlation coefficient is –

$$r = \frac{N\Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{[N\Sigma X^2 - (\Sigma X)^2][N\Sigma Y^2 - (\Sigma Y)^2]}}$$

here, N = No. of candidates = 5

$$\therefore r = \frac{5 * 28780 - (365 * 385)}{\sqrt{(5 * 27375 - 365^2)(5 * 28780 - 385^2)}}$$

$$\therefore r = 3375/3385.41 \approx 0.997 \Rightarrow \boxed{r = 0.997}$$

Here, the value is positive which suggests that as aptitude score ↑, the job performance rating also ↑.

→ As, the value is extremely close to +1, which signifies a very strong, nearly perfect +ve linear relationship between two variables.

→ The result suggests that the aptitude test is an excellent predictor of first-year job performance.

HBM25062

6. A marketing firm tested the effectiveness of a new advertisement campaign by asking two independent groups of consumers (Group X and Group Y) to rank 7 different features of the product (A through G) based on their perceived importance.

Feature	Group X Rank (R1)	Group Y Rank (R2)	d = R1-R2	d^2
A	3	3	0	0
B	5	4	1	1
C	6	5	1	1
D	1	1	0	0
E	4	7	-3	9
F	7	5	2	4
G	1	2	-1	1
				$\Sigma d^2 = 16$

1. Calculate the Spearman's Rank Correlation Coefficient
2. Would the marketing firm be justified in proceeding with a campaign that emphasizes the top-ranked features, assuming both groups reflect the general target market? Why or why not?

Q.1)

→ The spearman's Rank correlation Coefficient measures the degree of agreement between the rankings given by group X & group Y.

$$\therefore r_s = 1 - \frac{6 \Sigma d^2}{N(N^2-1)} = 1 - \frac{6 * 16}{7(49-1)}$$

$$\boxed{\therefore r \approx 0.714}$$

Q.2)

→ The calculated spearman's rank correlation coefficient is $r_s \approx +0.714$. This represents strong +ve correlation, indicating that two independent groups have high degree of agreement in their perception of product of feature importance

→ So, The marketing firm would be justified in proceeding with a compaign that emphasizes the top-ranked features.

↳ strong agreement → Strong +ve correlation

↳ Confidence in findings → since both groups are assumed to reflect the general target market, the high level of agreement provides the firm with confidence that the features ~~ind~~ identified as most important will resonate with the wider consumer base

↳ optimal resource allocation → both groups ranke higuly is the most effective & least risky strategy for advertising compaign

$\boxed{\text{HBM25062}}$

7. A recruitment agency wants to assess the relationship between the score a candidate receives on a Pre-Interview Aptitude and their eventual First-Year Job Performance Rating out of 100. They collected data for five candidates.

Candidate	Aptitude Test Score (X)	Job Performance Rating (Y)	X^2	Y^2	XY
1	65	70	4225	4900	4550
2	75	78	5625	6084	5850
3	80	85	6400	7225	6800
4	55	60	3025	3600	3300
5	90	92	8100	8464	8280
	$\Sigma X = 365$	$\Sigma Y = 385$	$\Sigma X^2 = 27375$	$\Sigma Y^2 = 30273$	$\Sigma XY = 28780$

1. Calculate the Pearson's Correlation Coefficient between the Aptitude Test Score and the Job Performance Rating
2. Interpret the calculated value of r. What does this result suggest about the usefulness of the aptitude test for predicting job performance?
3. If a new candidate scores 60 on the Aptitude Test, what would be a reasonable *statistical expectation* for their Job Performance Rating, based on your calculated correlation?

Q.1) Pearson's correlation coefficient $= r = \dfrac{N\Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{[N\Sigma X^2 - (\Sigma X)^2][N\Sigma Y^2 - (\Sigma Y)^2]}}$

here, N = no. of candidates $\boxed{N=5}$

$$\therefore r = \frac{5 * 28780 - (\overset{365}{\cancel{27375}}) * (\overset{385}{\cancel{30273}})}{\sqrt{(5 * 27375 - (365)^2)(5 * 30273 - (385)^2)}}$$

$$\therefore r = \frac{3375}{3385.41} \approx 0.997 \Rightarrow \boxed{r \approx 0.997}$$

Q.2) → Interpretation!

→ This is an extremely strong +ve correlation, as value is very close to +1, indicating the aptitude test score (X) & the job performance rating (Y) move together in a highly predictable linear fashion.

→ This result suggests the aptitude test is an excellent & highly useful predictor of first-year job performance. Candidates who scored high on test are overwhelmingly likely to perform well on the job, & candidates who ~~s~~ score low are likely to perform poorly. The test is strong proxy for future job success.

Q.7

Q.3) statistical expectation of new candidate (score of 60)

→ by linear regression eqⁿ ⇒ $\hat{Y} = a + bX$

where, $\hat{Y}$ is the performance rating.

$$b = \frac{N \Sigma XY - (\Sigma X)(\Sigma Y)}{N \Sigma X^2 - (\Sigma X)^2}$$

$$\therefore b = \frac{3375}{3650} \approx 0.9247$$

Now, $a = \bar{Y} - b\bar{X}$ ... (a = Y-intercept)

$$\bar{X} = \frac{\Sigma X}{N} = \frac{365}{5} = 73$$

$$\bar{Y} = \frac{\Sigma Y}{N} = \frac{385}{5} = 77$$

$$\therefore a = 77 - (0.9247) \times 73 \approx 9.4969$$

hence, $\hat{Y} = 9.4969 + 0.9247\,(X)$

$$\therefore \hat{Y} = 9.4969 + 0.9247\,(60) \approx 64.98$$

$$\boxed{\therefore \hat{Y} \approx 65}$$

→ A reasonable statistical expectation for a candidate who scores 60 on the Aptitude test is a job performance rating of approximately 65.

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8. A financial analyst wants to determine if the risk (measured by the variance) of two different investment portfolios, Portfolio A and Portfolio B, is significantly different.

They collected the following data on the daily returns (in percentage points) over a sample period:

Statistic	Portfolio A	Portfolio B
Sample Size (n)	$n_A = 21$	$n_B = 16$
Sample Variance (s^2)	$s_A^2 = 9.8$	$s_B^2 = 4.2$

The analyst wants to test the hypothesis that the variance of Portfolio A is significantly greater than the variance of Portfolio B, using a 5% level of significance.

→ State the hypothesis!

Null hypothesis (H_0): variances are equal $\Rightarrow \sigma_A^2 = \sigma_B^2$

Alternative hypothesis (H_a): $(Var)_A > (Var)_B \Rightarrow \sigma_A^2 > \sigma_B^2$

→ To determine test statistic & critical value using F-ratio.

$$F = \frac{(\text{Sample variance})_A}{(\text{Sample variance})_B} = \frac{S_A^2}{S_B^2}$$

here, $S_A^2 = 9.8$

$S_B^2 = 4.2$

Sample sizes $\Rightarrow$ $n_A = 21 \rightarrow$ DOF A (df_1) $= n_A - 1 = 20$

$n_B = 16 \rightarrow$ DOF B (df_2) $= n_B - 1 = 15$

& level of significance (α) = 5% = 0.05

here, from tabular data, $F_{critical}(0.05, 20, 15) \approx 2.33$

→ Calculation for F-ratio!

$$F_{calculated} = \frac{S_A^2}{S_B^2} = \frac{9.8}{4.2} \approx 2.333$$

Here, $\boxed{F_{calculate}(2.333) > F_{critical}(2.33)}$ hence, null hypothesis rejected.

→ There is sufficient evidence at 5% level of significance to conclude that the variance (risk) of portfolio A is significantly greater than the variance (risk) of portfolio B.

HBM25062

9. A manufacturing plant produces screws that must have an average length of 10.0 cm to meet quality control standards. A quality inspector takes a random sample of 15 screws and measures their lengths.

The sample results are:

- Sample Mean: = 9.85cm
- Sample Standard Deviation: 0.5 cm

The inspector wants to know if the sample mean significantly differs from the required standard of 10 cm, using a 1% level of significance (alpha = 0.01).

→ ① State the hypothesis!

Ho : The true avg. length of screws is equal to the standard

$H_0 : \mu = 10.0$ cm

H₁ : The true avg length of screws is different from the standard

$H_1 : \mu \neq 10.0$ cm

→ ② To determine critical value for t.

here, sample mean $(\bar{x}) = 9.85$ cm

population mean $(\mu_0) = 10.0$ cm

sample std. deviation $(s) = 0.5$ cm

sample size $(n) = 15$

$DoF = n-1 = 15-1 = 14$

level of significance $(\alpha) = 0.01$ (Two-tailed)

$$t_{calculated} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$\therefore t_{calculated} = \frac{9.85 - 10.00}{0.5/\sqrt{15}}$$

$$\boxed{\therefore t_{calculated} \approx -1.163}$$

→ ③ Critical value:

for a two tailed test with $\alpha = 0.01$ & $df = 14$, we look up

$t_{\alpha/2}$ $(t_{0.005, 14})$ ⇒ $\boxed{t_{critical} \approx \pm 2.977}$

here, reject Ho if $\boxed{|t_{calculated}| > t_{critical}}$ ⇒ $\boxed{1.163 < 2.977}$

since, $\boxed{|t_{calculated}| < t_{critical}}$ the calculated t-statistic does not fall into rejection region. We fail to reject the null hypothesis (Ho)

HBM25062

10. A company produces precision bolts that must have an average breaking strength of 150 kg. The population standard deviation for the breaking strength is 5 kg, which is known from extensive historical data.

A quality control engineer selects a random sample of 36 bolts and finds the sample mean breaking strength to be 148.5 kg.

The engineer wants to determine if the current batch of bolts has a significantly different mean breaking strength from the standard of 150 kg, using a 1% level of significance

→ ① state the hypothesis !

H_0 : $\mu = 150$ kg (Avg. Strength = standard)

H_1 : $\mu \neq 150$ kg (Avg. strength ≠ standard)

→ ② Given data:

Sample mean $(\bar{x}) = 148.5$ kg

Population mean $(\mu_0) = 150$ kg

Population std. deviation $(\sigma) = 5$ kg

sample size $(n) = 36$

level of significance $(\alpha) = 0.01$ (two-tailed)

as size ≥ 30, we will → use Z test.

$$Z_{(calculated)} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

$$\therefore Z_{(calculated)} = \frac{148.5 - 150}{5/\sqrt{36}}$$

$$\boxed{\therefore Z_{(calculated)} \approx -1.80}$$

→ ③ critical value !

For two-tailed test with $\alpha = 0.01$, significance level is split into two tails ($\alpha/2 = 0.005$).

$\therefore$ Z corresponds to the area of $(1 - 0.005) = 0.995$

$$\boxed{\therefore Z_{critical} \approx \pm 2.576}$$

→ ④ Make a decision:

Reject H_0, if $\boxed{|Z_{calculated}| > Z_{critical} \Rightarrow 1.80 < 2.576}$

Since, $\boxed{|Z_{calculated}| < Z_{critical}}$, the calculated Z-statistic does not fall into the rejection region. We fail to reject the null hypothesis (H_0).

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11. A retailer wants to know if there is a relationship between a customer's Gender and their Preferred Shopping Channel (In-store vs. Online) for a particular product category. They surveyed a random sample of 200 customers and collected the following observed frequencies:

Observed Frequencies (O)	In-store	Online	Row total
Male	40	60	100
Female	30	70	100
Col. total	70	130	200

→ Step ①: State the hypothesis

Ho : There is no relationship between gender & preferred shopping

H1 : There is relationship between gender & preferred shopping

→ step ②: Calculate marginal totals & expected frequencies (E)

$$E = \frac{\text{Row total} \times \text{Column Total}}{\text{Grand total}}$$

E	In-store	online
Male	35	65
female	35	65

$E_{11} = \frac{100 \times 70}{200} = \boxed{35}$ $E_{12} = \frac{100 \times 130}{200} = \boxed{65}$

$E_{21} = \frac{100 \times 70}{200} = \boxed{35}$ $E_{22} = \frac{100 \times 130}{200} = \boxed{65}$

→ step ③: To calculate chi-square test statistic (χ^2)

$$\chi^2 = \Sigma\left(\frac{(O-E)^2}{E}\right)$$

χ^2	In-store	online
male	0.714	0.385
female	0.714	0.385

$$\chi^2 \text{ calculated} = \Sigma \frac{(O-E)^2}{E}$$

$$\boxed{\approx 2.198}$$

→ step ④: critical value → assuming level of significance (α) at 5%.

DOF : df = (Row − 1) (Column − 1) = (2−1) * (2−1) = 1

$\boxed{\therefore \chi \text{ critical } (0.05, 1) \approx 3.841.}$

Reject Ho, if $\chi^2_{\text{calculated}} > \chi^2_{\text{critical}}$ but here, $\chi^2_{\text{calculated}} < \chi^2_{\text{critical}}$.

hence, There is not enough statistical evidence to conclude that there is a significant relationship between a customer gender & preferred shopping channel at 5% level of significance

HBM25062

12. A call center manager wants to determine if incoming customer service requests are distributed uniformly across the five working days of the week (Monday through Friday). If the distribution is uniform, it helps in staffing decisions.

In a given week, the following number of requests were observed (O):

Day	Monday	Tuesday	Wednesday	Thursday	Friday	Total
Observed Requests (O)	110	95	80	105	110	500
Expected Req (E)	100	100	100	100	100	500
$\frac{(O-E)^2}{E}$	1	0.25	4.00	0.25	1.00	6.50

→ Step 1:

Ho : customer service requests are distributed uniformly across the five working days

Ha : The customer service requests are not distributed uniformly across the five working days.

Total requests & expected frequency $(N) = 110+95+80+105+110$
$= 500.$

$\because E = \frac{N}{k} = \frac{500}{5} = 100$

→ Step 2: To find χ^2 calculated

$$\boxed{\chi^2_{calculated} = \sum \frac{(O-E)^2}{E} = 6.50}$$

→ Step 3: Determine critical value

Degree of freedom $df = k-1, = 5-1 = 4$

Now, $\chi^2_{critical}$ at $\alpha = 0.05$ & $df = 4$, $\boxed{\chi^2_{critical}(0.05, 4) \approx 9.488}$

→ Step 4: Reject Ho, if $\chi^2_{calculated} > \chi^2_{critical}$

here, $6.50 < 9.488$, at 5% level of significance, there is insufficient evidence to conclude that the customer service requests are not distributed uniformly across the five working days

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13. A high school teacher wants to determine the linear relationship between the Number of Hours a Student Spends Studying for a Test (X) and the student's final Exam Score (Y) out of 100.

Data was collected from a random sample of 5 students:

Student	Study Hours (X)	Exam Score (Y)	X^2	XY
1	2	55	4	110
2	4	65	16	260
3	5	70	25	350
4	7	85	49	595
5	9	95	81	855

$\Sigma X = 27$
$\Sigma Y = 370$
$\Sigma X^2 = 175$
$\Sigma XY = 2170$

Predict the expected exam score for a student who studies for 6 hours.

→ step ①: To determine Regression eqⁿ → $\hat{Y} = a + bX$

$$\text{slope } b = \frac{N\Sigma XY - (\Sigma X)(\Sigma Y)}{N(\Sigma X^2) - (\Sigma X)^2}$$

$$\therefore b = \frac{5(2170) - (27)(370)}{5(175) - (27)^2} = \frac{10850 - 9990}{875 - 729}$$

$$\boxed{\therefore b = \frac{860}{146} \approx 5.8904}$$

→ To find → Y-intercept (a)

$$a = \bar{Y} - b\bar{X} \Rightarrow \bar{X} = \frac{\Sigma X}{N} = \frac{27}{5} = 5.4$$

$$\bar{Y} = \frac{\Sigma Y}{N} = \frac{370}{5} = 74$$

$$\therefore a = 74 - (5.8904) * 5.4$$

$$\boxed{\therefore a \approx 42.1918}$$

∴ Regression eqⁿ is $\boxed{\hat{Y} = 42.1918 + 5.8904X}$

→ To predict the expected exam score:- $\boxed{\text{at } X = 6}$

$$\hat{Y} = 42.1918 + 5.8904 * (6)$$

$\boxed{\therefore \hat{Y} \approx 77.53}$ ⇒ Here, the expected exam score for a student who studies for 6 hours is approx. 77.53

HBM25062

14. An agricultural research team is testing a new, proprietary fertilizer blend to see if it significantly increases the crop yield of a specific wheat variety. To control for soil variability and climate, they use a paired design:

The harvest yields (in bushels per acre) are recorded below:

Plot	Standard Fertilizer (X1)	New Blend (X2)	$d = X_2 - X_1$	d^2
1	58	61	3	9
2	65	64	-1	1
3	70	73	3	9
4	62	68	6	36
5	55	60	5	25
6	75	74	-1	1
7	60	65	5	25
8	68	69	1	1
9	59	62	3	9
10	68	71	3	9
			$\Sigma d = 27$	$\Sigma d^2 = 125$

→ $N = 10$, Now, to find t-value

A) the mean diff, $\bar{d} = \frac{\Sigma d}{N} = \frac{27}{10} = 2.7$

B) Std. deviation of differences (sd)

$$S_d = \sqrt{\frac{\Sigma d^2 - \frac{(\Sigma d)^2}{N}}{N-1}} = \sqrt{\frac{125 - \frac{(27)^2}{10}}{9}}$$

$\boxed{\therefore S_d \approx 2.4060}$

C) Calculate the t-value ⇒ $t_{calculated} = \frac{\bar{d}}{S_d/\sqrt{N}} = \frac{2.7}{2.4060/\sqrt{10}}$

$\boxed{\therefore t_{calculated} \simeq 3.549}$

D) Critical value of t at $\alpha = 0.05$ & $df = N-1 = 9$

$\boxed{t_{critical}(0.05, 9) \simeq 1.833}$

Q14

Here,

Reject Ho, if $\boxed{t_{calculated} > t_{critical}}$

$\boxed{\& \quad 3.549 > 1.833}$

Since, the $t_{calculated}$ is greater than the critical value, we reject the null hypothesis (H_0).

Hence, the new proprietory fertilizer blend significantly increases the crop yield compared to the standard fertilizer at the 5% level of significance.